

ME 205: Machine Drawing

Course Code	ME 205
Course Name	Machine Drawing
L-T-P-C	1-0-4-3
Prerequisites	Graphics for Design
Intended for	B.Tech.
Distribution	Core
Approval	3rd Ad-hoc Senate (Nov 2012)

Course Contents

- **Introduction**
Introduction to Engineering design process and drawings. Drawing standards. Computer aided drafting and use of software packages for engineering drawings
- **Detachable Fasteners**
Screw threads, approximate and conventional representations; Specifications; Threaded fasteners: Types, forms, standard, and specifications; Drawing of temporary connections; Foundation bolts; Locking Devices: Classification, principles of operation, standard types and their proportions; Shaft Couplings: Common types, standard proportions for some couplings; Pipe Joints, common pipe connections, Cotter and Knuckle Joint
- **Permanent Fastenings**
Rivets: Standard forms and proportions; Riveted Joints: Common types of joints, terminology, proportions and representation; Welds: Types of welds and welded joints, edge preparation, specifications, and representation of welds on drawings.
- **Assembly drawings**
Assembly drawings with sectioning and bill of materials. Assemblies involving machine elements like shafts, couplings, bearing, pulleys, gears, belts, brackets. Detailed part drawings from assembly drawings. Engine mechanisms-assembly. Machine Tool drawings including jigs and fixtures.
- **Production drawings**
Limits, fits, and tolerances of size and form; Types and grade, use of tolerance tables and specification of tolerances, form and cumulative tolerances, tolerance dimensioning, general tolerances; Surface quality symbols, terminology and representation on drawings, correlation of tolerances and surface quality with manufacturing techniques.
- **Schematics, process and instrumentation diagrams**
- **Structural drawings**
examples for reading and interpretation

Text and Reference Books

1. French, T. E., Vierch, C. J., and Foster, R. J., **Engineering Drawing and Graphic Technology**, 14th Edition, McGraw-Hill, 1993.
2. Giesecke, F. E, and Lockhart, S.D, **Technical Drawing**, 13th Edition, Prentice-Hall, 2008
3. Sideswar, N., **Machine Drawing**, McGraw-Hill, 2004.
4. Lakshminarayanan, V., and Mathur, M. L., **Text Book of Machine Drawing (with Computer Graphics)**, 12th Ed, Jain Brothers, 2007.
5. Narayana K.L., Kannaiah, P., and Venkata Reddy K, **Machine Drawing**, 3rd Edition., New Age International Publishers, 2006.
6. Johan K. C., **Text Book of Machine Drawing**, PHI Learning Pvt, 2009.
7. SP 46: 1988 Engineering Drawing Practice for Schools and Colleges, Bureau of Indian Standards, 1988.